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ACRME Hard Constraints Reference

Classification: Principal Cloud Architect — Placement Engine Design

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Status: Production Ready

Overview

This document consolidates all **Hard Constraints (HC-1 through HC-10)** governing regional placement decisions in the **Azure Capacity Reservation Management Engine (ACRME)**. Hard constraints are **binary pass/fail gates** — any region failing a hard constraint is excluded from scoring entirely, not penalized with a lower score.

Enforcement Architecture

Hard constraints are enforced at two stages in the placement decision pipeline:

Stage 1 — Pre-filtering (before candidate set formation): - **HC-8** GEOGRAPHY_CONTAINMENT - **HC-9** STANDARD_REGION_ONLY

Stage 2 — Hard Constraint Gate (after geography/class filtering, before scoring): - **HC-1** REGION_SEPARATION - **HC-2** CAPACITY_FLOOR - **HC-3** QUOTA_FLOOR - **HC-4** DR_SEPARATION_CLASS - **HC-5** ZONE_AVAILABILITY - **HC-6** DR_COVERAGE_FLOOR - **HC-7** DR_FLOOR_INTEGRITY - **HC-10** CROSS_GEO_EXTENSION_PATH_APPROVED

Regions surviving both stages enter the **scoring pipeline** where soft objectives (PS_Prod, PS_NonProd, PS_DR) rank them.

Source Documents

- **HC-1 through HC-7:** multi_region_placement_design.md §27 (Hard Constraints)
 - **HC-8, HC-9, HC-10:** acrme_production_readiness_review_and_architecture.md §27 (Production-Ready Region Selection)
 - **Design rationale:** design_change_summary.md (Decisions D1–D9)
-

Part 1 — Hard Constraint Definitions

HC-1: REGION_SEPARATION [UPDATED]

Category: Isolation

Source: Multi-Region Placement Design §27; Decision D8

Status: Updated in Pass 2 — NonProd/DR co-location now permitted

Definition

NonProd_region Prod_region

DR_region Prod_region

[Constraint DR_region NonProd_region REMOVED - NonProd and DR may share a region]

Rationale NonProd/DR co-location is permitted to allow **DR overflow capacity reuse** from the NonProd CRG. This change (Decision D8) enables the engine to leverage unused NonProd capacity as a DR buffer, reducing the minimum region count from 4 to 3.

With 3 regions:

Prod is isolated; NonProd and DR both draw from the remaining 2 regions (they may land on the same region or on different ones — determined by HC-6 and PS score).

With 4 regions:

Prod eliminates 1; NonProd selects from 3; DR may share with NonProd or use the remaining regions — whichever satisfies HC-6 and maximises PS_DR.

Implementation

- Prod region is excluded from NonProd candidate set
- Prod region is excluded from DR candidate set
- NonProd region is **not** excluded from DR candidate set (co-location allowed)

- CVAL (alias for NonProd) and Prod must not share a region
- CVAL and DR may share a region only under the approved policy (HC-1 update per D8)

Validation Rules (VR)

- **VR-1:** Prod and DR must not share a region
- **VR-2:** CVAL and Prod must not share a region
- **VR-3:** CVAL and DR may share a region only under the approved policy

Evidence Tag [Derived] from Azure Well-Architected Framework (Reliability pillar)

HC-2: CAPACITY_FLOOR

Category: Capacity

Source: Multi-Region Placement Design §27

Status: Stable

Definition

CR headroom in candidate region `minimum_reservation_units`

where:

`minimum_reservation_units = 2 × requested_vm_count × SKU_vCPU_count` (default multiplier: 2)

CR headroom = `total_free_slots` (in the target CRG type: Prod/NonProd/DR)

Rationale Ensures sufficient capacity buffer exists in the region before placement. The **2× multiplier** provides headroom for growth and prevents placement into near-exhaustion regions.

Implementation

- Read `RegionalSnapshot.prod_crg_free_slots`, `nonprod_crg_free_slots`, or `dr_crg_free_slots` (per environment type)
- Compare against `minimum_reservation_units` (configurable per `PlacementPolicy`)
- **Regions at capacity floor are excluded from scoring entirely**

Configuration

- **Default multiplier:** 2
- **Configurable via:** `PlacementPolicy.capacity_floor_multiplier`
- **Overridable per geography or SKU class**

Evidence Tag [Undocumented - architectural judgement]

HC-3: QUOTA_FLOOR [UPDATED]

Category: Quota

Source: Multi-Region Placement Design §27; Design Change Summary QG-3

Status: Updated in Pass 2 — now reads from Quota Groups, not per-subscription `QuotaRecord`

Definition

For Prod placement in region R:

REJECT if: $\text{Prod_Group_headroom}(R) < \text{requested_vm_count} \times \text{vCPU_per_instance}$

For NonProd placement in region R:

REJECT if: $\text{nonprod_headroom}(R) < \text{requested_vm_count} \times \text{vCPU_per_instance}$
(nonprod_headroom uses effective_nonprod_ceiling - floor-adjusted, not raw group_limit)

For DR placement in region R:

REJECT if: $\text{dr_headroom}(R) < \text{target_dr_qty} \times \text{vCPU_per_instance}$
(target_dr_qty = customer_prod_vm_count × dr_ratio)

Data Source

- **Primary:** QuotaGroup entity (Cosmos DB), synced to RegionalSnapshot (Redis)
- **Fields:**
 - $\text{Prod_Group_headroom}(R)$ = Prod group available quota in region R
 - $\text{nonprod_headroom}(R)$ = NonProd+DR group available quota **minus DR floor** (see HC-7)
 - $\text{dr_headroom}(R)$ = DR portion of NonProd+DR group quota
- **Legacy field:** QuotaRecord.deployment_headroom is superseded for placement checks; retained for per-subscription quota monitoring only

Formulas Prod Group Headroom:

$\text{Prod_Group_headroom}(R) = \text{Prod_Group_Limit}(R) - \text{Prod_Group_Used}(R)$

where:

$\text{Prod_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_vcpu} / \text{base_subscription_quota_limit})$
 $\text{Prod_Group_Used}(R) = \sum (\text{prod_crg.quantity} \times \text{vCPU})$ for all Prod CRGs in region R

NonProd Headroom (effective ceiling):

$\text{nonprod_headroom}(R) = \text{effective_nonprod_ceiling}(R) - \text{nonprod_quota_used}(R)$

where:

$\text{effective_nonprod_ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$
 $\text{NonProd_DR_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_vcpu} / \text{base_subscription_quota_limit})$
 $\text{DR_Floor_vCPU}(R) = \text{potential_dr_demand}(R) \times \text{vCPU} \times \text{dr_ratio_max}$
 $\text{nonprod_quota_used}(R) = \sum (\text{nonprod_crg.quantity} \times \text{vCPU})$ for all NonProd CRGs in region R

DR Headroom:

$\text{dr_headroom}(R) = \text{DR_Floor_vCPU}(R) - \text{dr_quota_used}(R)$

where:

$\text{dr_quota_used}(R) = \sum (\text{dr_crg.quantity} \times \text{vCPU})$ for all DR CRGs in region R

Validation Sequence

1. **HC-3 (Quota Floor):** Check raw headroom availability
2. **HC-7 (DR Floor Integrity):** Check floor compliance (applies only to NonProd placements)

POC Blocker B-1: Azure Quota Groups functionality must be GA-available in target tenant/region. If `Microsoft.Capacity/resourceProviders/.../groupQuotas` returns 404, escalate to Azure Support. **POC-30** gates this.

Evidence Tag [Documented] — Azure Quota Groups Preview API documentation

HC-4: DR_SEPARATION_CLASS

Category: Resilience

Source: Multi-Region Placement Design §27

Status: Stable (non-paired regions only)

Definition

DR_region must have `Separation_Class(DR_region, Prod_region) = HIGH`

Rationale Ensures non-correlated failure domains between Prod and DR regions. This constraint applies **only to non-paired regions** (e.g., standalone configuration with 3-4 explicitly chosen regions). Paired regions (e.g., East US / West US) are assumed to have HIGH separation by Azure design.

Separation Class Matrix

Prod Region	DR Region	Separation Class	Eligible?
East US	West US	HIGH	Yes (paired)
East US	Central US	MEDIUM	No
East US	West Europe	HIGH	Yes (cross-geo)
UAE North	Saudi Arabia Central	MEDIUM	No (same geography)
UAE North	Belgium Central	HIGH	Yes (cross-geo extension)

Implementation

- Read `SeparationClassRegistry` (static config or Azure metadata service)
- For non-paired regions, validate `Separation_Class(DR, Prod) = HIGH`
- Paired regions bypass this check (assumed HIGH by default)

Evidence Tag [Undocumented - architectural judgement]

HC-5: ZONE_AVAILABILITY

Category: Resilience

Source: Multi-Region Placement Design §27

Status: Stable

Definition

Candidate region must have 2 physical Availability Zones

Rationale Single-zone regions are excluded for all non-dev environments. Multi-zone deployment is a **Well-Architected Framework Reliability pillar** requirement.

Implementation

- Read `RegionalSnapshot.zone_count` (populated from Azure metadata)
- Reject regions where `zone_count < 2`
- **Applies to:** Prod, CVAL, DR placements
- **Exemption:** Dev environments may bypass this constraint (configurable via `PlacementPolicy.allow_single_`

Examples

Region	Zone Count	Eligible?
East US	3	Yes
West Europe	3	Yes
Central India	3	Yes
Brazil South	1	No
West US 3	3	Yes

Evidence Tag [Undocumented - architectural judgement]

HC-6: DR_COVERAGE_FLOOR [NEW]

Category: DR Capacity

Source: Multi-Region Placement Design §27; Design Change Summary CR/CRG-1; Decision D8

Status: New in Pass 2 — operationalises NonProd/DR co-location

Definition

REJECT DR placement in region R if combined DR + NonProd overflow capacity is insufficient to absorb the new customer's DR demand:

$$\text{dr_crg_free_slots}(R) + \text{nonprod_crg_effective_free}(R) < \text{customer_requested_dr_slots}$$

where:

$$\text{customer_requested_dr_slots} = \text{prod_vm_count} \times \text{dr_ratio_max}$$

$$\text{nonprod_crg_effective_free} = \text{nonprod_crg_free_slots} - \text{nonprod_crg_dr_overflow_reserve}$$

$$\text{dr_ratio_max} = 0.40 \quad (\text{policy constant} - \text{upper bound for DR-to-Prod ratio})$$

Rationale Since DR and NonProd may share a region (HC-1 constraint removed per D8), the engine must verify that the **combined capacity pool** can absorb the customer's DR need — either from the DR CRG directly, or from the NonProd CRG's reserved overflow headroom.

This constraint operationalises HC-1 co-location at the capacity selection level.

Data Requirements Per-CRG-type snapshot fields (added in Pass 2): - `nonprod_crg_free_slots(R)` — available NonProd CR capacity - `nonprod_crg_effective_free(R)` — NonProd capacity minus DR overflow reserve - `nonprod_crg_dr_overflow_reserve(R)` — capacity earmarked for DR failover absorption - `dr_crg_free_slots(R)` — available DR CR capacity

Implementation Pseudocode

```
def validate_HC6_DR_coverage_floor(region, customer_prod_vm_count):
    snapshot = get_regional_snapshot(region)

    customer_requested_dr_slots = customer_prod_vm_count * DR_RATIO_MAX # 0.40

    combined_dr_capacity = (snapshot.dr_crg_free_slots +
                           snapshot.nonprod_crg_effective_free)

    if combined_dr_capacity < customer_requested_dr_slots:
        return REJECT, "HC-6 DR_COVERAGE_FLOOR: Insufficient combined DR+NonProd overflow capacity"

    return PASS
```

Policy Constants

- **dr_ratio_min:** 0.30 (minimum DR-to-Prod ratio)
- **dr_ratio_max:** 0.40 (maximum DR-to-Prod ratio, used in HC-6 floor calculation)
- **dr_ratio_target:** 0.35 (recommended ratio)

Evidence Tag [Derived] from Decision D8 (NonProd/DR co-location)

HC-7: DR_FLOOR_INTEGRITY [NEW]

Category: Quota Protection

Source: Multi-Region Placement Design §27; Design Change Summary QG-4

Status: New in Pass 2 — enforces DR floor within Two-Group Quota Architecture

Definition

REJECT NonProd placement in region R if placing the requested quantity would push `nonprod_quota_used` above `effective_nonprod_ceiling`:

$$\text{nonprod_quota_used}(R) + (\text{requested_vm_count} \times \text{vCPU_per_instance}) > \text{effective_nonprod_ceiling}(R)$$

where:

$$\text{effective_nonprod_ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$$

Rationale Prevents NonProd allocation from **encroaching on the protected DR quota floor**. Within the Two-Group Quota Architecture, the NonProd+DR group has a shared limit; the DR floor reserves a portion for DR use only.

Evaluation Order

1. **HC-3 (Quota Floor):** Check raw headroom (`nonprod_headroom` `requested`)
2. **HC-7 (DR Floor Integrity):** Check floor compliance (`used` + `requested` `effective_ceiling`)

HC-3 checks availability; HC-7 checks compliance.

Formulas Effective NonProd Ceiling:

`effective_nonprod_ceiling(R)` = `NonProd_DR_Group_Limit(R)` - `DR_Floor_vCPU(R)`

DR Floor (per region):

`DR_Floor_vCPU(R)` = `potential_dr_demand(R)` × `vCPU_per_instance` × `dr_ratio_max`

where:

`potential_dr_demand(R)` = Σ `prod_crg.quantity` for all Prod CRGs that could fail over to region R
`dr_ratio_max` = 0.40

POC Example (GP-06 Topology) Inputs: - Prod demand: 80 VMs × 2 vCPU = 160 vCPU - Base subscription quota limit: 200 vCPU

Group Limits: - **Prod Group:** $200 \times 1.28 = 256$ vCPU (28% emergency headroom) - **NonProd+DR Group:** $200 \times 1.28 = 256$ vCPU

DR Floor Calculation:

`DR_Floor_vCPU` = 80 VMs × 2 vCPU × 0.40 = 64 vCPU

Effective NonProd Ceiling:

`effective_nonprod_ceiling` = 256 - 64 = 192 vCPU

Enforcement:

If `nonprod_quota_used` = 150 vCPU:

New NonProd request for 60 vCPU × 2 = 120 vCPU
 $150 + 120 = 270 > 192$ REJECT (HC-7 violation)

If `nonprod_quota_used` = 100 vCPU:

New NonProd request for 40 vCPU × 2 = 80 vCPU
 $100 + 80 = 180 \leq 192$ PASS

POC Blocker B-2: Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim) must be validated. **POC-31** measures release latency; confirm < 5 min for Tier RTO compliance.

Evidence Tag [Derived] from Two-Group Quota Architecture design

HC-8: GEOGRAPHY_CONTAINMENT [IMPLIED]

Category: Geography

Source: Production Readiness Review §27

Status: Implied constraint (no standalone definition block)

Definition

In Scenario 1 (geography-based placement), derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography.

Rationale When a customer supplies an Azure geography (e.g., “US”, “Europe”, “Middle East”) rather than a specific region, the engine derives the Prod anchor via `argmax(PS_Prod)` over Standard Capacity Regions **within that geography only**. This constraint ensures the derived Prod region respects the customer's geographic preference.

Enforcement Stage Stage 1 — **Pre-filtering** (before candidate set formation)

Validation Rules (VR)

- **VR-4:** Derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography

Implementation

```
def get_prod_candidates(geography):
    all_regions = get_all_azure_regions()

    # Stage 1: Geography containment (HC-8)
    geo_filtered = [r for r in all_regions
                    if r.geography == geography and
                    r.region_class == "Standard"]

    # Stage 2: Apply HC-1..HC-7, HC-10
    hc_filtered = apply_hard_constraints(geo_filtered)

    # Stage 3: Score and rank
    return argmax(hc_filtered, key=lambda r: PS_Prod(r))
```

Examples

Customer Geography	Eligible Regions	Ineligible (HC-8)
US	East US, West US, Central US, ...	West Europe, UK South, ...
Europe	West Europe, North Europe, UK South, ...	East US, Australia East, ...
Middle East	UAE North, Saudi Arabia Central	East US, Belgium Central*

*Belgium Central becomes eligible only via **HC-10 Cross-Geo Extension Path** for DR, not for Prod.

Evidence Tag [Undocumented - architectural judgement]

HC-9: STANDARD_REGION_ONLY [NEW]

Category: Region Class

Source: Production Readiness Review §27

Status: New gate for production-ready region classification

Definition

Automated placement, scoring, recommendation, and all environment assignments (Prod, CVAL, DR) must use Standard Capacity Regions only.

Rationale Restricted Capacity Regions (sovereign clouds, government regions, regions with SKU limitations) are excluded from automated placement. Their exclusion is enforced at **Stage 1 of the eligibility decision tree**, not as a scoring penalty.

Exception deployments (customer explicitly requests a Restricted region) proceed via the **Scenario 2 exception path** only, requiring operator approval.

Region Classification Standard Capacity Regions: - Commercial Azure public cloud regions with full SKU availability - No sovereign cloud restrictions - No special approval required for CRG creation - Examples: East US, West Europe, Australia East, Japan East, Brazil South

Restricted Capacity Regions: - Azure Government (US DoD, US Gov) - Azure China (21Vianet) - Regions with limited SKU availability (flagged in Azure metadata) - Regions requiring special tenant approval (sovereign cloud onboarding)

Enforcement Stage Stage 1 — Pre-filtering (before candidate set formation)

Validation Rules (VR)

- **VR-5:** All automated placement paths (Scenario 1 geography-based, Prod derivation, NonProd/DR selection) must use Standard Capacity Regions only
- **VR-6:** CVAL and DR must not use Restricted Capacity Regions under any condition, including exception deployments

Implementation

```
def get_eligible_regions_stage1(geography, customer_request):
    all_regions = get_all_azure_regions()

    # HC-9: Standard regions only (Stage 1 pre-filter)
    standard_regions = [r for r in all_regions if r.region_class == "Standard"]

    # If customer explicitly requests a Restricted region + Scenario 2 exception path
    if customer_request.region_override and customer_request.region_override.region_class == "Res
        return handle_exception_deployment(customer_request)

    # HC-8: Geography containment
    geo_filtered = [r for r in standard_regions if r.geography == geography]

    return geo_filtered
```

Exception Deployment Workflow (Scenario 2) **Trigger:** Customer explicitly requests a Restricted Capacity Region

Workflow: 1. Operator receives request via exception deployment queue 2. Validate customer has required tenant permissions (sovereign cloud registration) 3. Validate SKU availability in target Restricted region 4. Manual approval gate (ops team review) 5. If approved: Create Prod CRG in Restricted region; **CVAL and DR must still use Standard regions** (VR-6) 6. Record exception in `OperationRecord` with approval metadata

Evidence Tag [Undocumented - architectural judgement]

HC-10: CROSS_GEO_EXTENSION_PATH_APPROVED [NEW]

Category: Cross-Geo DR

Source: Production Readiness Review §27

Status: New governance control for cross-geography DR assignments

Definition

Any DR assignment to a region outside the customer's chosen geography must match an explicitly enumerated Cross-Geo Extension path in the active `PlacementPolicy`.

Rationale Cross-geography DR introduces data residency, compliance, and latency considerations that require explicit approval. The engine does not autonomously select cross-geo DR regions; it only accepts pre-approved extension paths defined in the active `PlacementPolicy`.

Enforcement DR assignments to Standard Capacity Regions in a **different geography** are rejected if no approved extension path exists for the source geography.

Approved Cross-Geo Extension Paths (Example)

Source Geography	Approved DR Geographies	Primary DR Region	Rationale
Middle East	Europe	Belgium Central	Middle East has only 2 regions (UAE North, Saudi Arabia); insufficient for 3-region model
Australia	Asia Pacific	Singapore	Regional pair fallback
India	Asia Pacific	Singapore	Regional pair fallback

Middle East Three-Region Placement (Canonical Example) **Inputs:** - Customer geography: Middle East - Available Middle East Standard regions: UAE North, Saudi Arabia Central (2 only)

Placement: 1. **Prod:** $\text{argmax}(\text{PS_Prod})$ over {UAE North, Saudi Arabia Central} → e.g., UAE North 2. **CVAL/NonProd:** Remaining Middle East region → Saudi Arabia Central 3. **DR:** Cross-Geo Extension to **Belgium Central** (approved path)

Validation: - Belgium Central must pass **HC-1 through HC-10** including DR coverage floor (HC-6) - If Belgium Central fails any HC, the placement is **rejected with an ops alert** - The engine does **not** silently select any alternative outside the approved extension paths

Validation Rules (VR)

- **VR-6:** For Middle East geography, DR region must be Belgium Central via approved Cross-Geo Extension path
- **VR-11:** Cross-Geo Extension DR path must be explicitly approved in active PlacementPolicy

Implementation

```
def validate_HC10_cross_geo_extension(customer_geography, dr_region):
    policy = get_active_placement_policy()

    # Check if DR region is in same geography as customer
    if dr_region.geography == customer_geography:
        return PASS # Same-geo DR, HC-10 does not apply

    # Cross-geo DR → validate against approved extension paths
    approved_paths = policy.cross_geo_extension_paths.get(customer_geography, [])

    if dr_region not in approved_paths:
        return REJECT, f"HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED: DR region {dr_region.name} not

    return PASS
```

PlacementPolicy Configuration (Example)

```
{
  "policy_version": "2026-08-Q3",
  "cross_geo_extension_paths": {
    "Middle East": {
      "approved_dr_regions": ["Belgium Central"],
      "rationale": "Middle East has only 2 Standard regions; 3-region model requires cross-geo DR
    },
    "Australia": {
      "approved_dr_regions": ["Singapore", "Japan East"],
      "rationale": "Regional pair fallback for Australia East / Australia Southeast"
    }
  }
}
```

Evidence Tag [Undocumented - architectural judgement]

Part 2 — Hard Constraint Summary Table

HC	Name	Type	Enforcement Stage	Primary Impact	Updated in Pass
HC-1	REGION_SEPARATION	Isolation	Stage 2	Prod isolated from NonProd/DR; NonProd=DR co-location allowed	Pass 2 (D8)
HC-2	CAPACITY_FLOOR	Capacity	Stage 2	Minimum CR headroom required	Stable
HC-3	QUOTA_FLOOR	Quota	Stage 2	Minimum quota headroom per environment type (Prod/NonProd/DR)	Pass 2 (QG-3)
HC-4	DR_SEPARATION	Resiliency Class	Stage 2	DR region must have HIGH separation class from Prod	Stable
HC-5	ZONE_AVAILABILITY	Resiliency	Stage 2	Minimum 2 availability zones required	Stable
HC-6	DR_COVERAGED_FLOOR	Capacity	Stage 2	Combined DR+NonProd capacity must absorb DR demand	Pass 2 (D8)
HC-7	DR_FLOOR_INTEGRITY	On-GRA Protection	Stage 2	NonProd placement cannot encroach on DR quota floor	Pass 2 (QG-4)
HC-8	GEOGRAPHY_CONFINEMENT	Geography	Stage 1	Prod region must be within customer's chosen geography (Scenario 1)	Implied
HC-9	STANDARD_REGION_ONLY	Region Only	Stage 1	Only Standard Capacity Regions eligible for automated placement	PRR §27

HC	Name	Type	Enforcement Stage	Primary Impact	Updated in Pass
HC-10	CROSS_GEO_EXTENSION_PATH_APPROVED	Cross-geography DR requires explicit approval path			PRR §27

Part 3 — Enforcement Pipeline

Stage 1 — Pre-filtering (Eligibility Decision Tree)

Applied before candidate set formation:

```

All Azure Regions
↓
[HC-9: STANDARD_REGION_ONLY filter]
↓
Standard Capacity Regions
↓
[HC-8: GEOGRAPHY_CONTAINMENT filter] (if Scenario 1)
↓
Standard Regions in Customer Geography
↓
Candidate Set for Stage 2

```

Stage 2 — Hard Constraint Gate

Applied to each candidate region before scoring:

```

def apply_hard_constraints_stage2(region, environment_type, customer_context):
    """
    Returns: (PASS, None) or (REJECT, reason)
    """
    # HC-1: Region separation
    if not validate_HC1_region_separation(region, environment_type, customer_context):
        return REJECT, "HC-1 REGION_SEPARATION"

    # HC-2: Capacity floor
    if not validate_HC2_capacity_floor(region, environment_type):
        return REJECT, "HC-2 CAPACITY_FLOOR"

    # HC-3: Quota floor
    if not validate_HC3_quota_floor(region, environment_type, customer_context):
        return REJECT, "HC-3 QUOTA_FLOOR"

    # HC-4: DR separation class (DR only)
    if environment_type == "DR" and not validate_HC4_dr_separation_class(region, customer_context):
        return REJECT, "HC-4 DR_SEPARATION_CLASS"

```

```

# HC-5: Zone availability
if not validate_HC5_zone_availability(region):
    return REJECT, "HC-5 ZONE_AVAILABILITY"

# HC-6: DR coverage floor (DR only)
if environment_type == "DR" and not validate_HC6_dr_coverage_floor(region, customer_context):
    return REJECT, "HC-6 DR_COVERAGE_FLOOR"

# HC-7: DR floor integrity (NonProd only)
if environment_type == "NonProd" and not validate_HC7_dr_floor_integrity(region, customer_context):
    return REJECT, "HC-7 DR_FLOOR_INTEGRITY"

# HC-10: Cross-geo extension path (cross-geo DR only)
if environment_type == "DR" and region.geography != customer_context.geography:
    if not validate_HC10_cross_geo_extension(customer_context.geography, region):
        return REJECT, "HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED"

return PASS, None

```

Stage 3 — Scoring Pipeline

Regions surviving HC-1..HC-10 enter scoring:

- **Prod:** Ranked by $PS_Prod(r) = \text{argmax over survivors}$
- **NonProd:** Ranked by $PS_NonProd(r) = \text{argmax over survivors (excluding Prod region)}$
- **DR:** Ranked by $PS_DR(r) = \text{argmax over survivors (excluding Prod region; may include NonProd region per HC-1)}$

Part 4 — Validation Rules (VR) Cross-Reference

The Production Readiness Review defines **11 Validation Rules (VR-1..VR-11)** that enforce hard constraints:

VR	Description	Enforced HCs
VR-1	Prod and DR must not share a region	HC-1
VR-2	CVAL and Prod must not share a region	HC-1
VR-3	CVAL and DR may share a region only under approved policy	HC-1 (D8 update)
VR-4	Derived Prod region must be within customer's chosen geography	HC-8
VR-5	All automated placement paths must use Standard Capacity Regions only	HC-9

VR	Description	Enforced HCs
VR-6	CVAL and DR must not use Restricted Capacity Regions under any condition	HC-9
VR-7	Standard region passes HC-1 through HC-10 → if all excluded, exhaustion error	HC-1..10
VR-8	For Middle East geography, DR region must be Belgium Central via approved cross-geo path	HC-10
VR-9	If Belgium Central fails HC-1..HC-10, block placement with ops alert	HC-1..10
VR-10	Restricted region requested by customer → Scenario 2 exception path only	HC-9
VR-11	Cross-Geo Extension DR path must be explicitly approved in active PlacementPolicy	HC-10

Part 5 — POC Blockers & Testing

Critical POC Blockers

ID	Issue	Blocked HCs	POC Gate	Resolution
B-1	Azure Quota Groups functionality in target tenant/region	HC-3, HC-7	POC-30	GA availability check; if 404 → escalate to Azure Support
B-2	Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim)	HC-7	POC-31	Measure release latency; confirm < 5 min for Tier RTO

POC Test Coverage

POC-01 through POC-10: Hard constraint validation (HC-1..HC-10)

POC-11: Middle East cross-geo DR (HC-10)

POC-30: Quota Groups API integration (HC-3, HC-7)

POC-31: Quota pool release latency (HC-7 emergency transfer dependency)

Part 6 — Configuration & Policy

PlacementPolicy Schema (Hard Constraint Section)

```
{
  "policy_version": "2026-08-Q3",
  "hard_constraints": {
    "capacity_floor_multiplier": 2,
    "dr_ratio_min": 0.30,
    "dr_ratio_max": 0.40,
    "dr_ratio_target": 0.35,
    "allow_single_zone_dev": false,
    "cross_geo_extension_paths": {
      "Middle East": {
        "approved_dr_regions": ["Belgium Central"],
        "rationale": "Middle East has only 2 Standard regions; 3-region model requires cross-geo"
      }
    },
    "separation_class_overrides": {}
  }
}
```

Policy Constants

Constant	Default Value	Configurable?	Affected HCs
capacity_floor_multiplier		Yes (per geography/SKU)	HC-2
dr_ratio_min	0.30	Yes	HC-6, HC-7
dr_ratio_max	0.40	Yes	HC-6, HC-7
dr_ratio_target	0.35	Yes	Scoring (not HC)
min_zone_count	2	Yes	HC-5
allow_single_zone_dev	false	Yes	HC-5
cross_geo_extension_paths		Yes	HC-10

Part 7 — Implementation Checklist

Engineering Backlog Cross-Reference

HC	Primary Epic	Stories	Tasks
HC-1..HC-8	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702, S0703, S0704, S0705	T070201
HC-9, HC-10	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702	T070202
HC-3, HC-7 (Quota Groups)	EPIC-03 (Quota Management)	S0310, S0311, S0314	T031001..T031006

HC	Primary Epic	Stories	Tasks
HC-6 (DR Coverage)	EPIC-06 (DR Management), EPIC-08 (Placement)	S0601, S0704	T060101, T070401

Pre-Deployment Validation

- ☐ HC-1..HC-10 validation functions implemented (unit tested)
- ☐ Stage 1 pre-filtering logic (HC-8, HC-9) validated
- ☐ Stage 2 hard constraint gate logic (HC-1..HC-7, HC-10) validated
- ☐ RegionalSnapshot schema extended with HC-6 fields (nonprod_crg_effective_free, dr_crg_free_slots)
- ☐ QuotaGroup integration (HC-3, HC-7) validated via POC-30
- ☐ Cross-Geo Extension paths configured in PlacementPolicy
- ☐ Middle East three-region placement validated (HC-10)
- ☐ Exception deployment workflow (HC-9 Restricted region override) tested
- ☐ HC rejection reasons logged to OperationRecord for audit
- ☐ Validation Rules VR-1..VR-11 traced to HC enforcement points

Appendix A — Decision Log Cross-Reference

Decision	HCs Affected	Change
D8	HC-1, HC-6	Removed DR_region NonProd_region constraint; added HC-6 to operationalise co-location
D1, D2	HC-3, HC-7	Introduced Two-Group Quota Architecture; HC-3 now reads from Quota Groups; HC-7 enforces DR floor
D9	HC-9, HC-10	Introduced Standard/Restricted region classification; added HC-9 and HC-10 governance gates

Appendix B — Evidence Tags

All hard constraints carry one of four evidence tags:

- **[Documented]** — Derived from Azure platform documentation (e.g., Quota Groups API)
- **[Derived]** — Logical consequence of a documented constraint or design decision
- **[Decided]** — Explicit design choice (traceable to Decision Log D1–D9)
- **[Undocumented — architectural judgement]** — Azure Well-Architected Framework best practice, not explicitly documented by Microsoft

Document Status: Production Ready

Next Review: After POC-30 and POC-31 completion (Quota Groups GA validation)